

Question 01

@A(-2, -5) to B(2, 5)

Find the zone,

$$dx = 2 - (-2) = 4$$

$$dy = 5 - (-5) = 10$$

→ 01

$|dy| > |dx| \rightarrow$ near to Y axis

$\therefore$ zone = 1

convert zone 1 to zone 0,

$$A(-2, -5) \rightarrow A(-5, -2)$$

$$B(2, 5) \rightarrow B(5, 2)$$

Now, apply Midpoint,

$$dx = 5 + 5 = 10$$

$$dy = 2 + 2 = 4$$

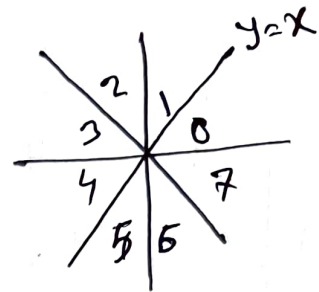
$$d_{init} = 2dy - dx = 2 \times 4 - 10 = -2$$

$$\Delta NE = 2dy - 2dx = (2 \times 4) - (2 \times 10) = -12$$

$$\Delta E = 2dy = 2 \times 4 = 8$$

$$NE \rightarrow d = d + \Delta NE \quad \left| \quad \begin{array}{l} NE \rightarrow A+1, B+1 \\ E \rightarrow A+1, B \end{array} \right.$$

$$E \rightarrow d = d + \Delta E$$



A	B	d	NE/E	$\Delta S/\text{updated } d$	Pixel(zone0)	Pixel(zone1)
-5	-2	-2	E	$d = -2 + 8 = 6$	$(-5, -2)$	$(-2, -5)$
-4	-2	6	NE	$d = 6 - 12 = -6$	$(-4, -2)$	$(-2, -4)$
-3	-1	-6	E	$d = -6 + 8 = 2$	$(-3, -1)$	$(-1, -3)$
-2	-1	2	NE	$d = 2 - 12 = -10$	$(-2, -1)$	$(-1, -2)$
-1	0	-10	E	$d = -10 + 8 = -2$	$(-1, 0)$	$(0, -1)$
0	0	-2	E	$d = -2 + 8 = 6$	$(0, 0)$	$(0, 0)$
1	0	6	NE	$d = 6 - 12 = -6$	$(1, 0)$	$(0, 1)$
2	1	-6	E	$d = -6 + 8 = 2$	$(2, 1)$	$(1, 2)$
3	1	2	NE	$d = 2 - 12 = -10$	$(3, 1)$	$(1, 3)$
4	2	-10	E	$d = -10 + 8 = -2$	$(4, 2)$	$(2, 4)$
5	2	-2	E	$d = -2 + 8 = 6$	$(5, 2)$	$(2, 5)$

⑥ $m = \frac{5+5}{2+2} = \frac{10}{4} = \frac{5}{2} = 2.5 > 1$

Now, $A_{k+1} = A_k + 1/m$ | $1/m = 1/2.5 = 0.4$

$B_{k+1} = B_k + 1$

B	A	round A	Pixel
-5	-2	-2	$(-2, -5)$
-4	-1.6	-2	$(-2, -4)$
-3	-1.2	-1	$(-1, -3)$
-2	-0.8	-1	$(-1, -2)$
-1	-0.4	0	$(0, -1)$
0	0	0	$(0, 0)$
1	0.4	0	$(0, 1)$
2	0.8	1	$(1, 2)$
3	1.2	1	$(1, 3)$

4	1.6	2	(2, 4)
5	2	2	(2, 5)

Question 02

① Total pixel = Width x Height
 $= 2560 \times 1440$
 $= 3686400 \text{ pixels}$

② Time taken = $\frac{1}{\text{FPS}} \times 1000$
 $= \left(\frac{1}{75} \times 1000 \right) \text{ ms}$
 $= \cancel{13.33 \text{ ms}}$
 $= \frac{40}{3} \text{ ms}$

③ Pixel per second = Total pixel x FPS
 $= (3686400 \times 75) \text{ pixels s}^{-1}$
 $= 276480000 \text{ pixels s}^{-1}$

GPU can render,
Pixel per second = $N \times \text{Time taken}$
 $= 75000 \times \cancel{13.33} \frac{40}{3} \text{ ms}$
 $= 1000000 \text{ pixels s}^{-1}$

So, GPU can not render one entire frame within the required time to maintain 75 fps.